

ASME IDETC/CIE 2026 • AUG 23–26

Heading to Houston!

Six Design Research Collective papers at the Hilton Americas–Houston.

**MON
24**

9:10–10:30 CT
DTM 01-01

DesignBench

9:10–10:30 CT
DAC-02-01

Structure Shapes Coordination: An Empirical Evaluation of Organizational Design Principles in Multi-Agent Engineering Design

2:10–3:50 CT
DEC-01

See One, Do One, Teach AI: Supporting Mastery and Self-Efficacy Through AI-as-a-Tutee

**TUE
25**

2:20–4:00 CT
DAC-03-02

Supervising Ralph Wiggum: Exploring a Metacognitive Co-Regulation Agentic AI Loop for Engineering Design

2:20–4:00 CT
DAC-03-02

Large Language Models as Complementary Agents for Dynamic Simulator-Driven Engineering Design Optimization

**WED
26**

8:00–9:40 CT
DTM 06-01

Designing for AI Adoption: A Multi-Scale Modeling Framework for Engineering Organizations

STUDENT PRESENTERS



Jessica Ezemba



Aslan Noorghasemi



Chan-Eui Song



Zeda Xu

SWIPE



DesignBench

DTM 01-01 · Benchmarking Design Research Progress



Presented by Zeda Xu

Wenzhuo Xu, Chan-Eui Song, Zachary Zdobinski, Zeda Xu, Mary Zhang, and Christopher McComb

ABSTRACT

Large language models (LLMs) have shown strong performance in mathematics, coding, and scientific question answering, motivating growing interest in their use for engineering design. However, most existing benchmarks evaluate static reasoning or one-shot generation, rather than the deeply iterative process by which designers interpret simulation feedback and revise artifacts over multiple steps. As a result, it remains unclear whether LLMs can reliably perform **closed-loop engineering design reasoning**. We introduce **DesignBench**, a benchmark designed specifically to evaluate this capability. DesignBench spans three design domains (i.e., truss optimization, planar linkage synthesis, and battery electrode design), each with a deterministic simulator that provide grounded feedback after every action. The benchmark harness represents design as search over **grammar-constrained modification trees**, enabling both reproducible evaluation and the generation of training traces for fine-tuning. Across these domains, we find that current frontier models can improve designs under feedback, but they do so unevenly. Performance reveals **systematic failure modes**, including a preference for local parametric edits over higher-leverage configuration changes, diminishing returns from long-context reasoning, and substantial dependence on feedback structure. These results position DesignBench as both a benchmark and a **training resource** for developing more reliable LLM-based design agents.

Structure Shapes Coordination: An Empirical Evaluation of Organizational Design Principles in Multi-Agent Engineering Design

DAC-02-01 • AI & ML for Design/Synthesis with Real-World Complexity



Presented by Jessica Ezemba

Jessica Ezemba, Christopher McComb, and Conrad Tucker

ABSTRACT

Multi-agent systems (MAS) and **organizational design (OD)** research have developed largely in parallel: MAS demonstrates promising automated capabilities across complex engineering tasks, while OD offers empirically grounded findings on how structure shapes coordination in human teams. Whether these bodies of knowledge meaningfully connect remains an open empirical question. Recent work proposed a taxonomy linking three MAS organizational structures (Orchestrated, Networked, and Sequential) and three operational methodologies (Iterative Feedback, Graph-Routed, and Staged Pipeline) to OD theory, yielding **eight valid combinations**, but stopped short of empirical evaluation. This paper presents the first implementation and evaluation of all eight combinations on a **multidisciplinary aircraft optimization** case study using NASA's Aviary framework. All eight converged to fuel burn rates between 6,166 and 6,352 kg, reaching a lighter-wing design region distinct from the 7,001 kg gradient-based reference optimum. OD-grounded metrics revealed measurably **distinct coordination regimes** independent of fuel outcome: Sequential combinations reduced coordination overhead at the cost of pipeline viscosity; Orchestrated combinations maintained stable delegation costs regardless of methodology pairing; and Networked combinations showed low convergence scores across all pairings, suggesting agents did not build on each other's contributions despite shared blackboard access. This work represents an initial empirical step toward understanding where **human organizational principles** transfer to MAS and where agent systems may require fundamentally different structures.

See One, Do One, Teach AI: Supporting Mastery and Self-Efficacy Through AI-as-a-Tutee

DEC-01 • AI in Design Education



Presented by Christopher McComb

Giovanni Natasha, Anshika Agarwal, Lydia Eckstein, Michael Melville, H. Elisabeth Ellington, Zach Mineroff, and Christopher McComb

ABSTRACT

The use of **generative AI tools** is increasingly common in engineering education, both formally and informally. Embracing these tools presents an opportunity to enhance student learning through novel pedagogical approaches, and many efforts have explored the use of AI agents to deliver personalized content and information for students. In contrast, this project investigates the impact of using a Large Language Model (LLM) as a “tutee” in a sophomore-level mechanical engineering course. Building on the “see one, do one, teach one” paradigm of mastery, this study explores whether teaching technical skills to an **AI tutee** can improve **student mastery** of core mechanical engineering concepts. This work evaluates performance on a well-established **concept inventory** across a group of students who repeatedly interact with an AI tutee, compared to a group with no such interaction. For students engaging with the AI tutee, we also capture and analyze **self-efficacy**, chat transcripts, and free response reflection questions. Preliminary results indicate the potential value of utilizing GenAI as a tutee. Ultimately, the aim of this research is to contribute **scalable teaching strategies** for large engineering classes and address disparities in student preparedness and backgrounds by providing a consistent and accessible mastery opportunity.

Supervising Ralph Wiggum: Exploring a Metacognitive Co-Regulation Agentic AI Loop for Engineering Design

DAC-03-02 · Novel AI or ML Frameworks for Design or Systems Science



Presented by Zeda Xu

Zeda Xu, Nikolas Martelaro, and Christopher McComb

ABSTRACT

The engineering design research community has studied **agentic AI systems** that use Large Language Model (LLM) agents to automate the engineering design process. However, these systems are prone to some of the same pathologies that plague humans. Just as human designers, LLM design agents can fixate on existing paradigms and fail to explore alternatives when solving design challenges, potentially leading to suboptimal solutions. In this work, we propose (1) a novel **Self-Regulation Loop (SRL)**, in which the Design Agent self-regulates and explicitly monitors its own metacognition, and (2) a novel **Co-Regulation Design Agentic Loop (CRDAL)**, in which a Metacognitive Co-Regulation Agent assists the Design Agent in metacognition to mitigate **design fixation**, thereby improving system performance for engineering design tasks. In the battery pack design problem examined here, we found that the novel SRL and CRDAL systems generate designs with **better performance**, without significantly increasing the computational cost, compared to a plain Ralph Wiggum Loop (RWL). Further, the novel CRDAL generates designs with significantly better performance than SRL. Also, we found that the CRDAL system navigated through the **latent design space** more effectively than both SRL and RWL. The proposed system architectures and findings of this work provide practical implications for future development of agentic AI systems for engineering design.

Large Language Models as Complementary Agents for Dynamic Simulator-Driven Engineering Design Optimization

DAC-03-02 · Novel AI or ML Frameworks for Design or Systems Science



Presented by Chan-Eui Song

Chan-Eui Song, Jeremy J. Michalek, and Christopher McComb

ABSTRACT

Large language models (LLMs) have emerged as potent tools for supporting engineering design. Unlike numerical optimization solvers, LLMs lack theoretical guarantees, but they have the unique potential to leverage **application-specific semantic information** (e.g., engineering knowledge), which may be particularly advantageous when optimizing black-box simulations, or in dynamic settings, where numerical solvers typically lack theoretical guarantees and/or scalability. Despite this potential, comparatively little is understood about how LLMs can contribute to solution search when simulator-defined design problems evolve during search and when different modes of LLM integration may lead to different optimization roles. In this study, we investigate when and how LLM agents can support **simulator-driven engineering design optimization** under static and **dynamic conditions**. We examine three levels of LLM integration in this setting: one-shot, **feedback-driven**, and **hybrid warm-start agents**. We evaluate these in the context of a lithium-ion battery pack design problem using the BatPaC simulator. We further examine the impact of a material price shock introduced during optimization. The results show that the feedback-driven approach can be competitive with Bayesian optimization and gradient methods at finding feasible designs and improving the objective function, especially under dynamic conditions. We also find that hybrid warm-start agents can achieve strong optimization performance by initializing conventional numerical optimization with LLM-generated starting points. These findings suggest that LLMs may serve as complementary components for semantic guidance in simulator-driven engineering optimization.

Designing for AI Adoption: A Multi-Scale Modeling Framework for Engineering Organizations

DTM 06-01 · Rethinking Design in the Age of Artificial Intelligence



Presented by Aslan Noorghasemi

Aslan Noorghasemi and Christopher McComb

ABSTRACT

Artificial intelligence (AI) systems are increasingly used to support engineering decision-making in engineering design, with a significant focus on the technical performance of the systems. However, adoption depends not only on technical performance but also on **social influence** within organizations. Existing approaches for predicting and understanding adoption often examine technical and social factors separately, making it difficult to understand how they jointly shape adoption dynamics. This paper develops a **multi-scale modeling framework** to facilitate the study of how engineers update opinions about AI tools through observed outcomes and peer interactions, and how these microfoundations influence organizational patterns. Using this framework, we analyze how **AI accuracy**, organizational structure, and initial sentiment toward AI shape adoption dynamics. Results show that adoption depends strongly on team size, number of teams, and baseline sentiment toward AI. Under negative initial sentiment, larger numbers of teams and larger team sizes require higher AI accuracy to achieve adoption. At moderate AI accuracy, adoption is usually dominated by **quiet adopters**, individuals who adopt AI but remain less socially influential. However, in large organizations with negative initial sentiment toward AI, **loud adopters**, who are more socially influential proponents of the AI, become a prominent driver of organizational adoption.